8.6

a. H_0: $\sigma_M^2 = \sigma_F^2$, H_a: $\sigma_M^2 \neq \sigma_F^2$

test statistic: $F = \frac{\widehat{\sigma_M^2}}{\widehat{\sigma_F^2}} = \frac{\frac{97161.9174}{573}}{12.024^2}$=1.1728

rejection region: $F > F_{0.025,573,419}$=1.196781

Since $F < F_{0.025,573,419}$ => Non-reject H_0, there is no significant proof that the wage variance between men and women are different.

b. H_0: $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$, H_a: $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$

test statistic: $F = \frac{\widehat{\sigma_{MARRIED}^2}}{\widehat{\sigma_{SINGLE}^2}} = \frac{\frac{100,703.0471}{595}}{\frac{56231.0382}{395}}$=1.1889

rejection region: $F > F_{0.05,595,395}$=1.164705

Since $F > F_{0.05,595,395}$ => Reject H_0, accept H_a. there is statistically significant that the wage variation for married group is larger than single group.

c. H_0: the error is Homoskedasticity, H_a: the error is Heteroskedasticity

test statistic NR^2=59.03~χ_4^2

Since 59.03>$\chi_{4,0.05}^2 = 9.487729$ => Reject H_0, accept H_a. the error is Heteroskedasticity

This doesn't provide evidence about the issue discussed in part (b) because the variable of XR8.6b doesn't include MARRIED.

d. H_0: the error is Homoskedasticity, H_a: the error is Heteroskedasticity

4 original explanatory variables, 2 square, 6 cross products

=>degree of freedom=12

test statistic NR^2=78.82~χ_{12}^2

Since 78.82>$\chi_{12,0.05}^2$=21.02607, Reject H_0, accept H_a. the error is Heteroskedasticity e.

gotten narrower: EXPER、METRO、FEMALE

gotten wider: intercept、EDUC

there is no inconsistency.

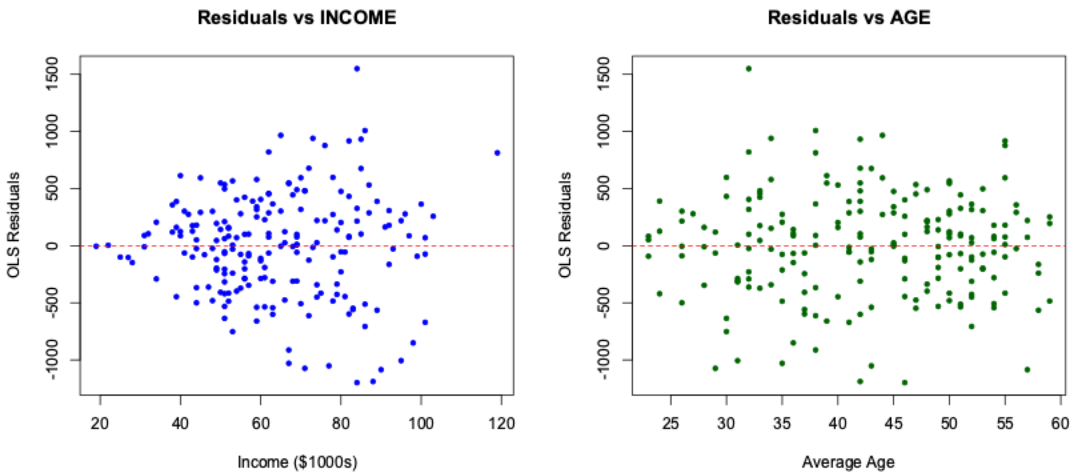
f. The results are completely compatible. Part (b) shows marital status affects wage *variance* (spread), while part (f) shows it does not affect the *mean* (average) wage due to t=1.0 (non-significant). A variable can easily increase wage volatility without changing the average, so there is no conflict.

8.16

a. Holding the two other variables constant, the 95% C.I. for KIDS

=>[-135.3298 -28.32302]. Add one or more kids in travel, MILES will be smaller.

b. We can observe that the residual for income is heteroskedasticity



c. $H_0: \sigma^2_{low} = \sigma^2_{high}, H_a: \sigma^2_{low} < \sigma^2_{high}$

```
> cat("高所得組 SSE:", sse_high, "\n")
高所得組 SSE: 27160654
> cat("低所得組 SSE:", sse_low, "\n")
低所得組 SSE: 8750040
> cat("F 檢定統計量:", f_stat, "\n")
F 檢定統計量: 3.104061
> cat("5% 臨界值:", f_crit, "\n")
5% 臨界值: 1.428617
```

Since $F > F_{86,0.05}$, Reject H_0, accept H_a. There exist Heteroskedasticity. When INCOME is higher, the variation will also be larger.

d. 95% C.I. using Robust Standard Errors => [-140.328781, -23.32406]

which is wider than the result in part (a). It makes sense because of Heteroskedasticity.

e. Conventional GLS and robust GLS have almost the same result.

It is great to assume $\sigma_i^2 = \sigma_i^2 INCOME_i^2$, compared the result with a and d, the C.I. is narrower due to Homoskedasticity.

```
GLS 傳統信賴區間 (e):
> print(gls_ci)
         2.5 %    97.5 %
kids -119.8945 -33.71808
> cat("\nGLS 強健信賴區間 (e):\n")

GLS 強健信賴區間 (e):
> print(robust_gls_ci["kids", ])
     2.5 %     97.5 %
-121.41339  -32.19919
```